

PROBLEMS

Week: 26 March / 26 Maart 2018

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

=====CHAPTER 5=====

P5.66 (a) $P_{\max} = 48.6 \text{ N}$ and $P_{\min} = 31.7 \text{ N}$ (b) Explanation (c) The block cannot slide up the wall. If $P < 62.7 \text{ N}$, it will slide down the wall.

P5.88

$$(a), (b) \quad T_1 = 2mg \sin \theta \qquad M = 3m \sin \theta \qquad T_2 = 3mg \sin \theta$$

$$(c), (d) \quad a = \frac{g \sin \theta}{1 + 2 \sin \theta}, \quad T_1 = 4mg \sin \theta \left(\frac{1 + \sin \theta}{1 + 2 \sin \theta} \right) \text{ and}$$

$$T_2 = 6mg \sin \theta \left(\frac{1 + \sin \theta}{1 + 2 \sin \theta} \right)$$

$$(e) \quad M_{\max} = 3m(\sin \theta + \mu_s \cos \theta).$$

$$(f) \quad M_{\min} = 3m(\sin \theta - \mu_s \cos \theta).$$

$$(g) \quad T_{2,\max} - T_{2,\min} = M_{\max} g - M_{\min} g = 6\mu_s mg \cos \theta$$

=====CHAPTER 6=====

OQ6.1 (a) $A > C = D > B = E = 0$. (b). (c).

OQ6.4 (a)

OQ6.5 (b)

OQ6.6 (a)

(b)

CQ6.2 The drag force is proportional to the speed squared and to the effective area of the falling object. At terminal velocity, the drag and gravity forces are in balance. When the parachute opens, its effective area increases greatly, causing the drag force to increase greatly. Because the drag and gravity forces are no longer in balance, the greater drag force causes the speed to decrease, causing the drag force to decrease until it and the force of gravity are in balance again.

CQ6.3

CQ6.4 (a)

(b)

CQ6.6 I would not accept that statement for two reasons. First, to be “beyond the pull of gravity,” one would have to be infinitely far away from all other matter. Second, astronauts in orbit are moving in a circular path. It is the gravitational pull of Earth on the astronauts that keeps them in orbit. In the space shuttle, just above the atmosphere, gravity is only slightly weaker than at the Earth’s surface. Gravity does its job most clearly on an orbiting spacecraft, because the craft feels no other forces and is in free fall.

CQ6.7

CQ6.9

P6.3 (a) $8.83 \times 10^{-8} \text{ N}$ inwards (b) $9.15 \times 10^{22} \text{ m/s}^2$ inwards

P6.8 Show the free-body diagram for this problem.

(a) $\vec{T} = T \sin \theta \hat{i} + T \cos \theta \hat{j} = (68.6 \text{ N}) \hat{i} + (784 \text{ N}) \hat{j}$

(b) $a_c = 0.857 \text{ m/s}^2$

P6.29

$$F = 3.01 \text{ N}$$

P6.45 (a) (1) The downward gravitational force

(2) The tension force in the string,
always directed toward the
center of the path

(b)

(c) 6.05 N

(d) 7.82 m/s

P6.50 (a) $v = \sqrt{Rg \tan 35.0^\circ} = \sqrt{(6.86 \text{ m/s}^2) R}$

(b) The mass is unnecessary.

(c) increasing the radius will make the required speed increase.

(d) When the radius increases, the period increases.

(e) On a larger circle, the ice cube’s speed is proportional to $\sqrt{R}$ but the distance it travels is proportional to R , so the time interval required is proportional to $R/\sqrt{R} = \sqrt{R}$.

P6.59 (a) $T = \sqrt{\frac{4\pi^2 R \mu_s}{g}}$

- (b) The gravitational and friction forces remain constant. (Static friction adjusts to support the weight.) The normal force increases. The person remains in motion with the wall.
- The gravitational force remains constant.
- (c) The normal and friction forces decrease. The person slides relative to the wall and downward into the pit.

P6.61 (a) $v_{\max} = \sqrt{\frac{Rg (\tan \theta + \mu_s)}{1 - \mu_s \tan \theta}}$

(b) $\mu_s = \tan \theta$.